SMARTSHELF - REPORT

AI-Powered Replenishment Assistant

AI CAPSTONE PROJECT – SKEMA – 2025/2026

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Table of Contents

Executive Summary 2
Business Problem and Project Motivation 2
Dataset and Pilot Scope 3
Data Preparation and Cleaning 3
Feature Engineering 4
Modeling Strategy and Baseline 4
  Model comparison 5
  Validation Strategy 5
Forcasting Results 5
  Walk-forward cross-validation — MAE 5
  Holdout results — 56 unseen days 6
Cost Model and Business Value 6
  Cost simulation results — 56 days, 3 stores, 100 products 7
Prototype and Application 7
Engineering and Reproducibility 8
Limitations and Future Work 8
Conclusion 8
References 9
Table of Figures 9

Executive Summary

SmartShelf is an AI-based replenishment decision-support system for retail stores. It uses five years of Walmart sales, calendar, and price data from the M5 dataset to forecast product demand, compare multiple models against realistic retail baselines, and translate forecast errors into simulated inventory cost in euros. The revised version addresses the jury's feedback by replacing the single-model approach with a comparison of LightGBM, XGBoost, and Prophet, evaluated with walk-forward cross-validation and a final 56-day holdout. LightGBM achieved the best overall performance, with MAE 3.35 and a simulated inventory cost of €40,137, compared with €74,577 for the seasonal naive baseline. This corresponds to €34,440 saved and a 46% reduction in inventory cost over 56 days across three stores and 100 products.

Figure 1: Key Results Summary

Business Problem and Project Motivation

Retail store managers make a replenishment decision every day: what to reorder, and how much. In many stores, this is still done using intuition or spreadsheet rules based on recent average sales. This decision logic is simple, but it ignores seasonality, price effects, events, and recurring weekly demand patterns. The result is unstable replenishment quality and unnecessary cost.

Two operational risks dominate this problem. The first is overstock, when predicted demand is higher than actual demand, creating excess inventory, holding cost, and tied-up capital. The second is stockout, when predicted demand is lower than actual demand, leading to empty shelves, lost sales, and customer churn. SmartShelf was designed to replace this reactive logic with machine-learning-based daily recommendations.

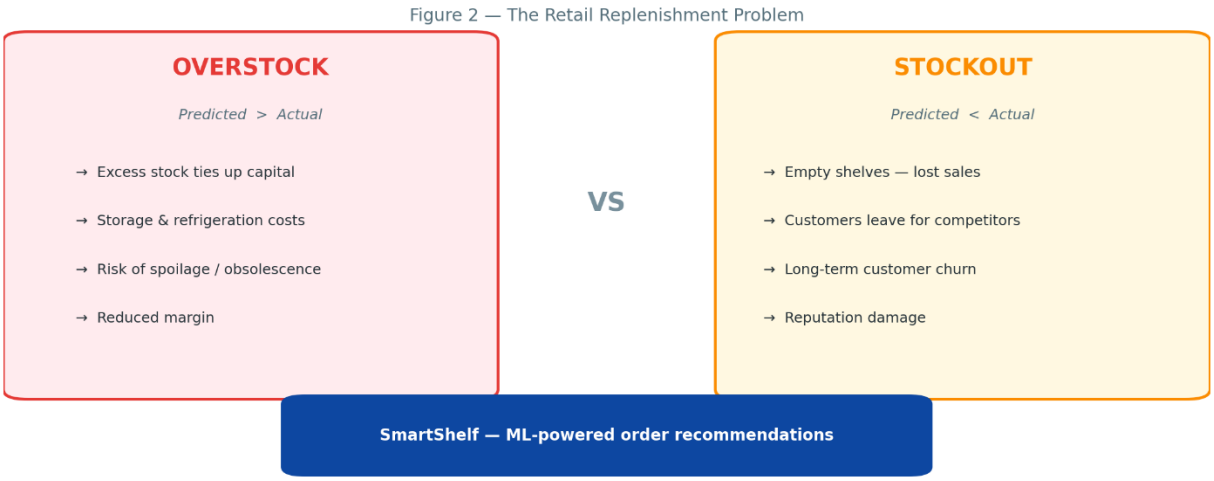

Figure 2: The Replineshment Problem

Dataset and Pilot Scope

The project is based on the M5 Forecasting dataset and uses three core sources: historical daily sales, calendar variables, and weekly item prices. To keep the system feasible on a standard laptop while preserving operational relevance, the pilot scope was limited to three stores and the top 100 products by total sales volume. This follows a Pareto principle: a relatively small number of products captures most of the revenue.

File	Content
sales_train_evaluation.csv	Daily unit sales — 1,941 days × 3,049 items, wide format
calendar.csv	Dates, events, holidays, SNAP benefit flags
sell_prices.csv	Weekly sell price per item per store — sparse

Scope element	Value
Stores	CA_1, TX_1, WI_1
Product selection	Top 100 by total sales volume
Dataset size	~582,000 rows
Period	2011–2016
Scale objective	Designed to scale to all 10 stores and 3,049 products

Data Preparation and Cleaning

Step	Operation
1	Filter sales to 3 stores and top 100 products by total sales volume
2	Melt wide format (1,941 day-columns) to long format (one row per day per item)
3	Merge calendar events and sell prices on item × store × week
4	Forward-fill then backward-fill missing sell prices per item-store group
5	Drop rows with no price data at all
6	Cast sales to int16 and sell_price to float32 to reduce memory

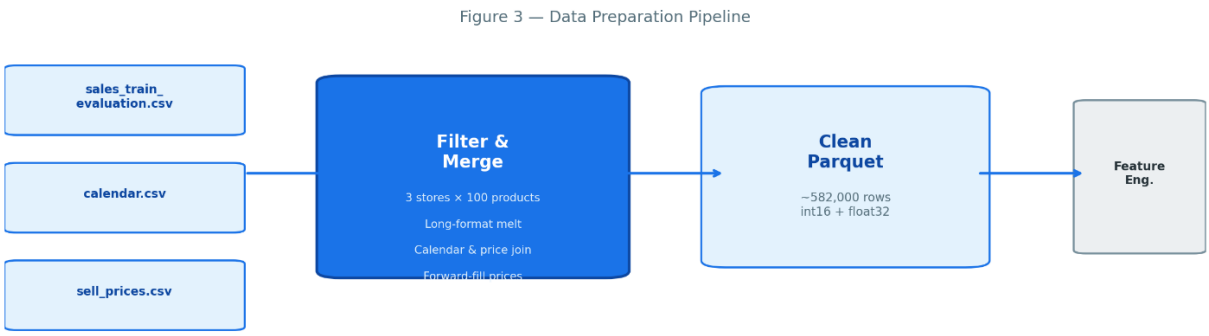

Figure 3: Data Preparation Pipeline

Feature Engineering

Family	Features	Purpose
Historical demand	lag_7, lag_14, lag_21, lag_28	Capture weekly and monthly sales patterns
Rolling statistics	rolling_mean_7/14/28, rolling_std_7/28, rolling_max_7/28	Trend, volatility, peak demand
Price signals	sell_price, price_change, price_vs_store_avg	Price elasticity and promotions
Calendar & events	day_of_week, is_weekend, is_month_end, has_event, snap_day, month, quarter	Seasonality and external demand drivers
Demand pressure	demand_7d, demand_28d, sales_velocity, zero_streak	Recent demand intensity and stockout history
Product identity	item_id, store_id, cat_id, dept_id, state_id	Store and category effects (label-encoded)

Total engineered features: 28

All categorical variables are label-encoded. No one-hot encoding is used, which keeps the feature space compact and matches the selected model family.

Modeling Strategy and Baseline

The first version of SmartShelf used a single LightGBM model with a Tweedie objective and a rolling 28-day average as the main baseline. This was changed after the defense. The revised version introduces multiple models, stronger baselines, and a chronological time-series validation design.

Model comparison

Model	Type	Objective	Why chosen
LightGBM	Gradient boosting	MAE	Fast, native categoricals, industry standard for M5
XGBoost	Gradient boosting	MAE	Alternative implementation for comparison
Prophet	Additive time series	Decomposition	Handles trend + weekly + yearly seasonality natively
Seasonal naive	Baseline	—	Predict same weekday 52 weeks ago — real retail standard
Rolling mean 28d	Baseline	—	28-day average — simple heuristic, kept for reference

Validation Strategy

Fold	Training data	Validation
1	Jan 2011 → Sep 2015	Oct 2015 (28 days)
2	Jan 2011 → Dec 2015	Jan 2016 (28 days)
3	Jan 2011 → Feb 2016	Mar 2016 (28 days)
Holdout	Jan 2011 → Mar 2016	Apr–May 2016 (56 days)

Forcasting Results

Walk-forward cross-validation — MAE

Model	CV MAE
LightGBM	3.35
XGBoost	3.38
Seasonal naive	6.28
Rolling mean 28d	8.31

Holdout results — 56 unseen days

Model	MAE	RMSE
LightGBM	3.35	5.38
XGBoost	3.37	5.43
Prophet	6.07	9.34
Seasonal naive	6.22	9.53
Rolling mean 28d	8.75	12.93

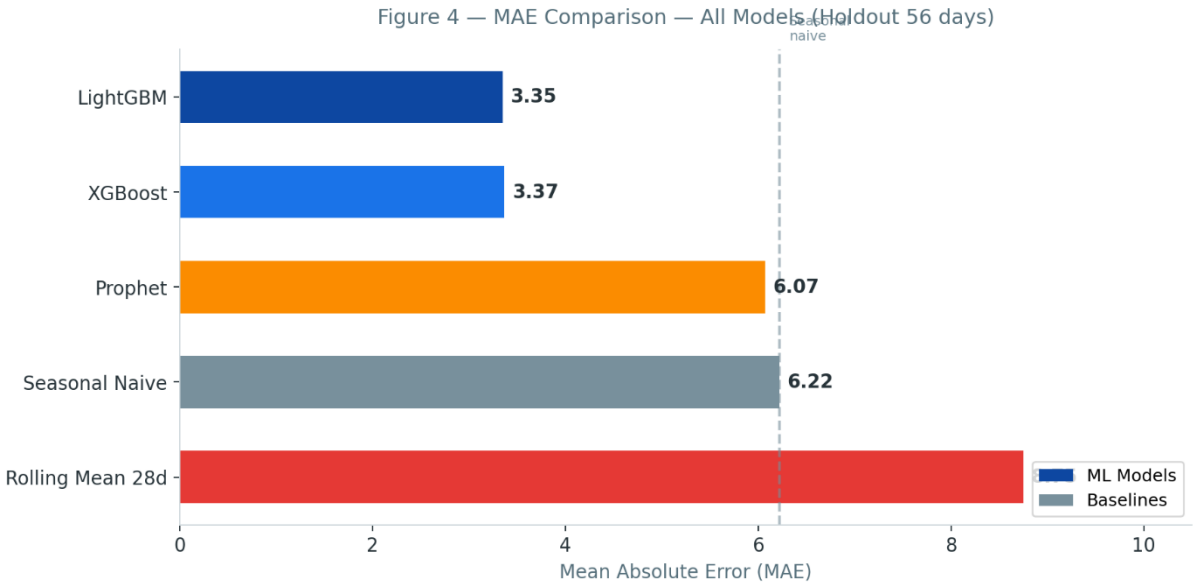

Figure 4: MAE Comparison

LightGBM achieves an MAE of 3.35, which is nearly 2x better than the seasonal naive baseline (6.22) and about 2.6x better than the rolling mean baseline (8.75).

Cost Model and Business Value

Cost type	Formula
Overstock cost	excess_units × sell_price × 0.068% per day
Stockout cost	missing_units × sell_price × 75%

Cost simulation results — 56 days, 3 stores, 100 products

Model	Simulated cost (€)
LightGBM	40,137
XGBoost	40,561
Prophet	74,491
Seasonal naive	74,577
Rolling mean 28d	102,798

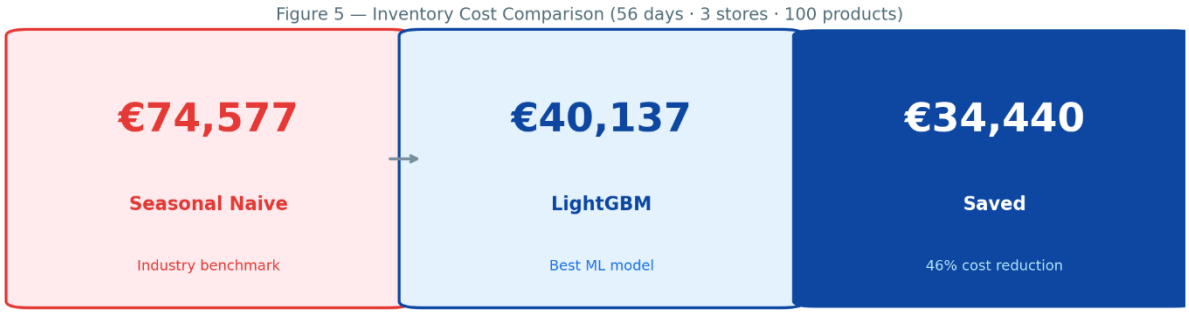

Figure 5: Cost Comparison

Prototype and Application

Page	What it shows
Order Assistant	Order quantity + stockout risk alert + SHAP explanation + 7-day forecast chart
Model Comparison	CV MAE per fold, holdout MAE/RMSE, error distribution histogram for all models
Cost Dashboard	Euro savings vs seasonal naive, by store and by category, daily cost over time
Model Insights	Feature importance chart, SHAP values, CV stability across folds

Engineering and Reproducibility

Aspect	Implementation
Language	Python 3.10+
Pipeline	Single command: python run_pipeline.py
Memory management	Store-by-store processing with PyArrow ParquetWriter
Model persistence	joblib for LightGBM and XGBoost, Prophet per-item
Reproducibility	Fixed random seed (seed=42), requirements.txt pinned
Version control	GitHub: github.com/MNadirZejli/SmartShelf_AICapstone
App deployment	Streamlit — streamlit run app/app.py

Limitations and Future Work

Current limitations	Future improvements
Cost rates are fixed constants from literature	Category-specific rates (perishables vs. durables)
Trained on 2011–2016 data — needs retraining for new patterns	Automated retraining triggered by data drift detection
Top-100 scope over-represents FOODS category	Scale to all 10 stores and 3,049 products
Prophet evaluated on only 20 sampled items	Full Prophet training across all items
No real stock data — stock level entered manually	Real-time POS system integration

Conclusion

SmartShelf is not just a forecasting model. It is a complete replenishment decision-support system that combines multiple time-series models, realistic retail baselines, walk-forward validation, euro-based cost simulation, and an interactive application. The final system achieves MAE 3.35, reduces simulated inventory cost to €40,137, and saves €34,440, corresponding to a 46% reduction compared with the seasonal naive baseline. SmartShelf shows how AI can move from prediction to operational retail decisions.

References

- Silver, E. A., Pyke, D. F., & Thomas, D. J. (1998). Inventory and Production Management in Supply Chains.
- ECR Europe. (2003). The Optimal Shelf Availability.

Table of Figures

Figure 1: Key Results Summary 2
Figure 2: The Replineshment Problem 2
Figure 3: Data Preparation Pipeline.....4
Figure 4: MAE Comparison.....6
Figure 5: Cost Comparison 7